SIM800 GSM / GPRS / SMS Add-on V2.0

Rev 1.2, Feb 2017

Contents

1、Overview....3
2、Feature....3
3、Specification....3
4、Hardware....3
5、Pinmap....3
6、Indicators....5
7、Keys....5
8、Extension....5
9、Note....6
10、RPI SIM800 operation instruction....6
11、Schematic....8

1、Overview

Raspberry PI SIM800 GSM/GPRS Add-on V2.0 is customized for Raspberry Pi interface based on SIM800 quad-band GSM/GPRS/BT module. AT commands, can be sent via the serial port on Raspberry Pi, thus functions such as dialing and answering calls, sending and receiving messages and surfing on line can be realized. Moreover, the module supports powering-on and resetting via software.

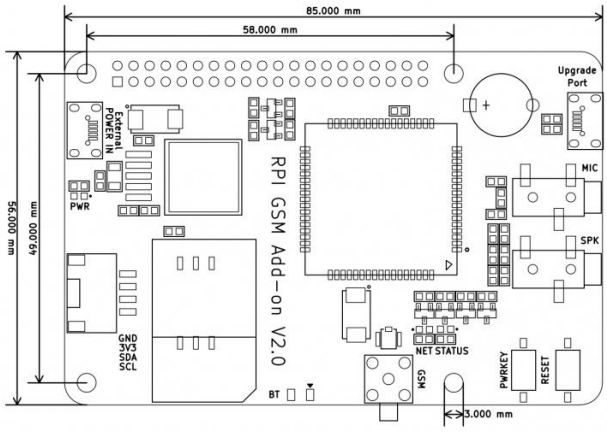
2、Feature

- Quad-Band 850/ 900/ 1800/ 1900 MHz
- GPRS multislot class 12
- Operation temperature: -40°C to +85 °C
- Stackable

3、Specification

- PCB size: 85mm X 56mm X 1.6mm
- Operation Level: Digital 3.3V DC
- Interface: UART
- Baud rate: 9600(default)

4、Hardware



5、Pinmap

Raspberry PI Pin N.O.	Pin name	Description
1	3.3V	
2	5V	
3	GPIO02	SDA
4	5V	
5	GPIO03	SCL
6	GND	
7	GPIO04	
8	GPIO14	GSM_DIN
9	GND	
10	GPIO15	GSM_DOUT
11	GPIO17	SIM800-PWRKEY
12	GPIO18	SIM800-RST
13	GPIO27	
14	GND	
15	GPIO22	
16	GPIO23	
17	3.3V	
18	GPIO24	
19	GPIO10	SPI_MOSI
20	GND	
21	GPIO09	SPI_MISO
22	GPIO25	
23	GPIO11	SPI_SCK
24	GPIO08	SPI_CE0
25	GND	
26	GPIO07	SPI_CE1
27	ID_SD	
28	ID_SC	
29	GPIO05	
30	GND	
31	GPIO06	
32	GPIO12	
33	GPIO13	

34	GND	
35	GPIO19	
36	GPIO16	
37	GPIO26	
38	GPIO20	
39	GND	
40	GPIO21	

6、Indicators

- PWR

  Power: When there is normal supply to the board, the indicator keeps on.

- 8STATUS

  SIM800 status: When SIM800 works normally, the indicator keeps on.

- NET

  SIM800 network status: used to indicate network status, the working status of the indicator is as below:

Status	SIM800 behavior
OFF	SIM800 does not work
64ms on/800ms off	SIM800 does not find network
64ms on/3000ms off	SIM800 registers the network
64ms on/300ms off	GPRS communicates

7、Keys

- PWRKEY

  SIM800 powering-on key: Keep pressing the key for more than 1 second, SIM800 will be powered on; keep pressing the key for more than 1 second after the board is powered on, SIM800 will be powered off.

- RST

  SIM800 resetting key: press the key, SIM800 will be reset.

8、Extension

- External POWER IN

If SIM800 will drop off while working, please connect an external at least 5V/2A power supply to this micro USB.

- Upgrade Port

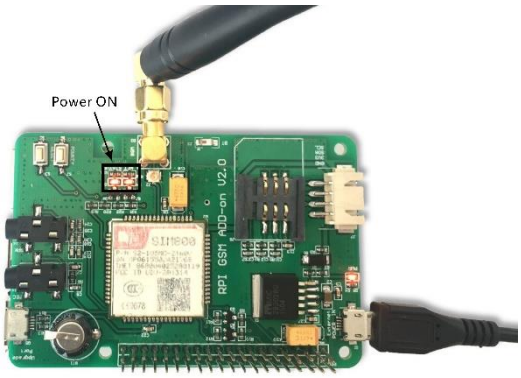
This is only for upgrading firmware of SIM800.

9、Note

The module does not support hot swapping.

10、RPI SIM800 operation instruction

Note: Firstly, you should key down the PWRKEY until the SIM800 is powered on.



Download the ArducamSIM800 library and put it in your /home/pi folder.

GitHub link: https://github.com/supprot/ArducamSIM800.git

Hardware:

If you are using a Raspberry Pi Model 1 A+/B+, Model 2/3 B or Zero then this add-on board will connect directly to the 40-pin GPIO.

If you are using a Model 1 A/B with a 28-pin GPIO connector, then you will have to connect the board to the Pi with ribbon cable or Dupont connectors as the composite video and audio connectors are in the way. Only the follow seven pins need to be connected:

Pin	Function
1	3.3V
2	5V
6	GND

8	TXD
10	RXD
11	SIM800 Power
12	SIM800 Reset

Software

This library has been tested with Python 3.4.2 running on a Raspberry Pi Model B (Rev.2) with Raspbian Jessie Lite (2016-05-27).

It depends on PySerial and Ben Croston's RPi. GPIO which can be installed (if not already) as follows:

Support for Pi /Pi2/ Pi Model B

sudo apt-get update

sudo apt-get install python3-rpi.gpio

sudo pip3 install pyserial

The file 'sms.py' is both the library and a working example if read/executed:

sudo python3 sms.py

Support for Pi3

There is now a device tree file called pi3-miniuart-bt which makes the Raspberry Pi 3 disable the Bluetooth and map pl011 UART on pins 14 and 15 as before.

- Step 1 - Install Raspbian Jessie onto a SD card and boot the Pi when connected to a network

  Login via terminal or desktop and shell

  Configure the system with:

  sudo raspi-config

  Expand filesystem and enable serial on advanced page, exit and reboot.

  Update the system with:

  sudo apt-get update

  sudo apt-get upgrade

- Step 2: Device Tree settings as below:

  Add device tree to /boot/config.txt to disable the Raspberry Pi 3 bluetooth.

  sudo nano /boot/config.txt

Add at the end of the file

dtoverlay=pi3-miniuart-bt

Exit the editor saving your changes and then:

sudo reboot

Enabling the Serial Console Rasbian Jessie after 18th March 2016 release

To enable the serial console, you need to edit the /boot/cmdline.txt file

sudo nano /boot/cmdline.txt

Change the file to the following:

dwc_otg.lpm_enable=0 console=tty1 console=serial0,115200 root=/dev/mmcblk0p2 rootfstype=ext4 elevator=deadline fsck.repair=yes rootwait

Exit and save your changes

sudo apt-get update

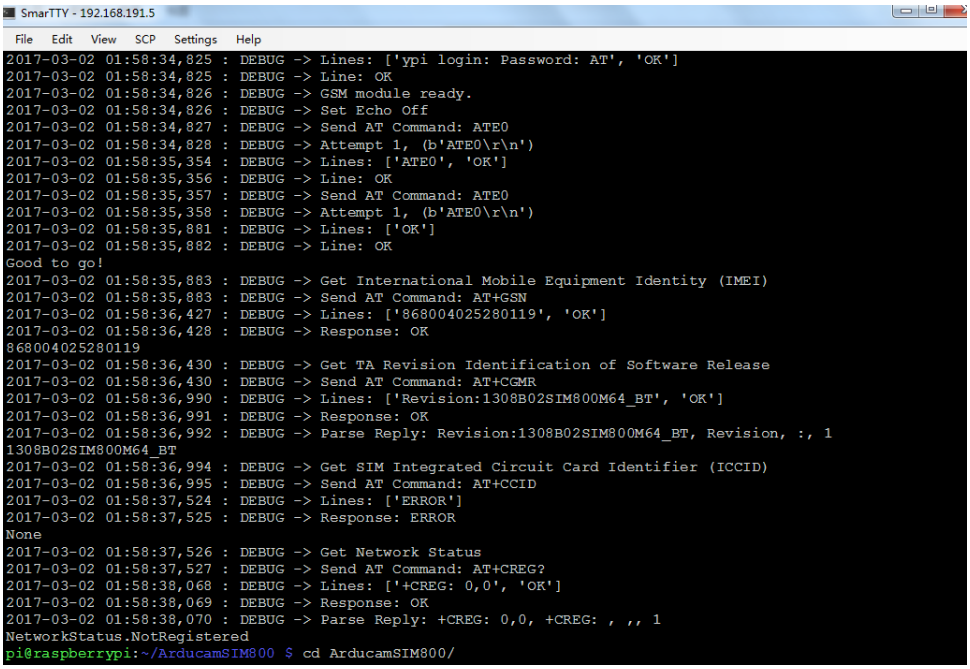
sudo apt-get install python3-rpi.gpio

sudo pip3 install pyserial

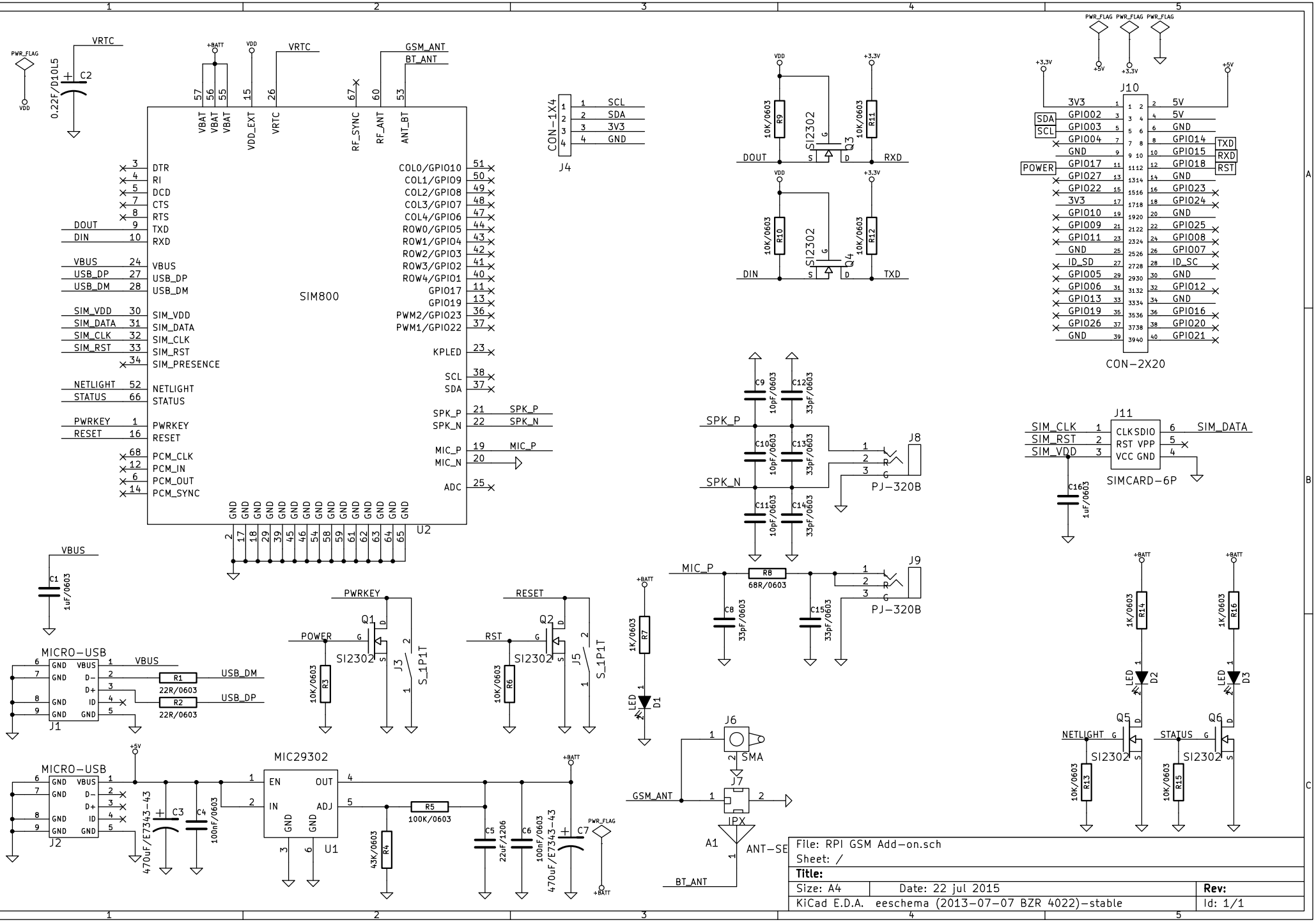
The file 'sms.py' is both the library and a working example if read/executed:

sudo python3 sms.py

Attach the demo running image:



11、Schematic


SIM800
U2
MIC29302
U1
CON-2X20
J10
SIMCARD-6P
J11
PJ-320B
J8
PJ-320B
J9
MICRO-USB
J1
MICRO-USB
J2
CON-1X4
J4
SMA
J6
IPX
J7
ANT-SE
A1
S_1P1T
J3
S_1P1T
J5
SI2302
Q1
SI2302
Q2
SI2302
Q3
SI2302
Q4
SI2302
Q5
SI2302
Q6
LED
D1
LED
D2
LED
D3
PWR_FLAG
VRTC
GSM_ANT
BT_ANT
VBUS
PWRKEY
RESET
POWER
RST
SPK_P
SPK_N
MIC_P
DOUT
DIN
RXD
TXD
NETLIGHT
STATUS
SDA
SCL
USB_DM
USB_DP
SIM_CLK
SIM_RST
SIM_VDD
SIM_DATA
ID_SD
ID_SC
GSM_ANT
BT_ANT
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Date: 22 jul 2015
Rev:
KiCad E.D.A. eeschema (2013-07-07 BZR 4022)-stable
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